

```

1 /*
2  * assembly.s
3  *
4  */
5
6 @ DO NOT EDIT
7 .syntax unified
8     .text
9     .global ASM_Main
10     .thumb_func
11
12 @ DO NOT EDIT
13 vectors:
14     .word 0x20002000
15     .word ASM_Main + 1
16
17 @ DO NOT EDIT label ASM_Main
18 ASM_Main:
19
20 @ Some code is given below for you to start with
21 LDR R0, RCC_BASE @ Enable clock for GPIOA and B by setting bit 17 and 18 in RCC_AHBENR
22 LDR R1, [R0, #0x14]
23 LDR R2, AHBENR_GPIOAB @ AHBENR_GPIOAB is defined under LITERALS at the end of the code
24 ORRS R1, R1, R2
25 STR R1, [R0, #0x14]
26
27 LDR R0, GPIOA_BASE @ Enable pull-up resistors for pushbuttons
28 MOVS R1, #0b01010101
29 STR R1, [R0, #0x0C]
30 LDR R1, GPIOB_BASE @ Set pins connected to LEDs to outputs
31 LDR R2, MODER_OUTPUT
32 STR R2, [R1, #0]
33 MOVS R2, #0 @ NOTE: R2 will be dedicated to holding the value on the LEDs
34
35 @ TODO: Add code, labels and logic for button checks and LED patterns
36
37 main_loop:
38 @start of student's code
39 @read the buttons
40 LDR R3, GPIOA_BASE
41 LDR R3, [R3, #0x10] @get the state of GPIOA->IDR
42
43 @check if BTN3 is pressed
44     MOVS R4, #8
45     ANDS R4, R4, R3
46     BEQ main_loop @if BTN3 is pressed, continue to next iteration
47
48 @check if BTN2 is pressed
49     MOVS R4, #4
50     ANDS R4, R4, R3
51     BNE skip_btn2 @if BTN2 is not pressed, skip the next instruction
52     MOVS R2, #0xAA @set the LED pattern to 0xAA
53     B write_leds
54 skip_btn2:
55
56 @check if BTN1 is pressed
57 LDR R5, LONG_DELAY_CNT @preemptively assume the delay is normal
58 MOVS R4, #2
59 ANDS R4, R4, R3
60     BNE skip_btn1 @skip the BTN1 special function
61 LDR R5, SHORT_DELAY_CNT @set the delay to 0.3s
62 skip_btn1: @normal delay
63
64 @check if BTN0 is pressed
65 MOVS R4, #1

```

```

66 ANDS R4, R4, R3
67 BNE skip_btn0 @skip the BTN0 special special function
68 MOVS R4, #2
69 ADD R2, R2, R4 @increment LEDs by 2
70 B completed_btn0
71 skip_btn0: @normal increment
72 MOVS R4, #1
73 ADDS R2, #1 @increment LEDs by 1
74 completed_btn0: @special increment
75
76 @applying delay
77 loop:
78 SUBS R5, #1 @decrement the counter
79 BNE loop @next iteration of the loop
80 @end of student's code
81
82 write_leds:
83 STR R2, [R1, #0x14]
84 B main_loop
85
86 @ LITERALS; DO NOT EDIT
87 .align
88 RCC_BASE: .word 0x40021000
89 AHBENR_GPIOAB: .word 0b11000000000000000000
90 GPIOA_BASE: .word 0x48000000
91 GPIOB_BASE: .word 0x48000400
92 MODER_OUTPUT: .word 0x5555
93
94 @ TODO: Add your own values for these delays
95 LONG_DELAY_CNT: .word 1400000 @0.7s*8MHz/4ClockCycles
96 SHORT_DELAY_CNT: .word 600000 @0.3s*8MHz/4ClockCycles
97

```